

Lab: Hashing Lab

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Passwords:

1. P@ssw0rd123
2. Summer2024!
3. letmein!

1. Part A—Compute hashes (CyberChef + Linux)

1a. P@ssw0rd123 plaintext hash

The screenshot shows the CyberChef web interface. On the left is a sidebar with a list of operations: Operations, SHA-256, SHA2, Favourites, Data format, Encryption / Encoding, Public Key, Arithmetic / Logic, Networking, Language, Utils, Date / Time, Extractors, Compression, Hashing, Code tidy, Forensics, Multimedia, Other, and Flow control. The main area is divided into three sections: 'Recipe', 'Input', and 'Output'. The 'Recipe' section contains three steps: MD5, SHA1, and SHA2. The 'Input' section shows the plaintext 'P@ssw0rd123'. The 'Output' section displays the resulting SHA256 hash: 'dc7f14585ecc259ed130cb0839d960733dd6701bf3acbc923f8dd1d281f8e20d'.

Operations	Recipe	Input
SHA-256	MD5	P@ssw0rd123
SHA2	SHA1	
Favourites	Rounds: 80	
Data format	SHA2	
Encryption / Encoding	Size: 256, Rounds: 64	
Public Key		
Arithmetic / Logic		
Networking		
Language		
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

1b. P@ssw0rd123 salt

Operations	Recipe	Input
SHA-256	MD5	321dr0vss@P
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

11

1

Raw Bytes

LF

Output

99b98a173c8ae3baa0acc4f938ca4fdbdb5c7b83c50ebab8af72cfe8df168dfc2

1c. Salted P@ssw0rd123 string

Operations	Recipe	Input
SHA-256	MD5	321dr0vss@PP@ssw0rd123
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

22

1

Raw Bytes

LF

Output

64d17f84ac6e46f2f3e8c7058fa830a8a4afe749fd3d1b26c5dfd48acec5f0e8

2a. Summer2024! Plaintext hash

Operations

SHA-256

SHA2

Favourites

Data format

Encryption / Encoding

Public Key

Arithmetic / Logic

Networking

Language

Utils

Date / Time

Extractors

Compression

Hashing

Code tidy

Forensics

Multimedia

Other

Flow control

Recipe

MD5

SHA1

Rounds80

SHA2

Size256

Rounds64

Input

Summer2024!

Output

4eade8587306e67f780a8524bcb9dc0ee0fb7605fdd5fa4578982f856c51f7ab

2b. Summer2024! Salt

Operations

SHA-256

SHA2

Favourites

Data format

Encryption / Encoding

Public Key

Arithmetic / Logic

Networking

Language

Utils

Date / Time

Extractors

Compression

Hashing

Code tidy

Forensics

Multimedia

Other

Flow control

Recipe

MD5

SHA1

Rounds80

SHA2

Size256

Rounds64

Input

I420zremmu5

Output

cf2765d589149619eb875a8a8879e2555e698ee7a38d030b63316a700ab7e049

2c. Salted Summer2024! String

Operations	Recipe	Input
SHA-256	MD5	!4202remuSSummer2024!
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

Output
9765c38ce8d7e99347fc6bac416047469c1c18b32074a41ca94c44ef415a56b5

3a. Letmein! Plaintext hash

Operations	Recipe	Input
SHA-256	MD5	letmein!
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

Output
248a2db30ff6703cbfa3271b81fb923bfec443e2241e034158a355f18644ab4d

3b. Letmein! Salt

Operations	Recipe	Input
SHA-256	MD5	!niemetel
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

Output
5f119378668f51bc5d35a813b8e86b5cb8da52430345084046b02dc53bfa8d91

3c. Salted Letmein! String

Operations	Recipe	Input
SHA-256	MD5	!niemetletmein!
SHA2	SHA1	
Favourites	Rounds	
Data format	80	
Encryption / Encoding	SHA2	
Public Key	Size	
Arithmetic / Logic	256	
Networking	Rounds	
Language	64	
Utils		
Date / Time		
Extractors		
Compression		
Hashing		
Code tidy		
Forensics		
Multimedia		
Other		
Flow control		

Output
04e3e1e71ce92fd6b679378303e22c019eaf605ab2d6d327edd810ceeb9e4b8e

4a. Linux P@ssw0rd123 Output:

```

root@JackLinux:~#
root@JackLinux:~# echo -n 'P@ssw0rd123' |md5sum
ccf5538dc31d435d6bab145c924041d8 -
root@JackLinux:~# echo -n 'P@ssw0rd123' |sha1sum
0f0d959bca569bf2b0a8bff3e2f1e88920ee7c5f -
root@JackLinux:~# echo -n 'P@ssw0rd123' |sha256sum
231ecc7d178da5f22983bc579599396d6c139a457987ae1ee0026d88432d6a72 -
root@JackLinux:~# echo -n 'P@ssw0rd123' |rev
321dr0wss@Proot@JackLinux:~#
root@JackLinux:~# echo -n '321dr0wss@P' |md5sum
8abccb25ed082a6382f412a915b282e2 -
root@JackLinux:~# echo -n '321dr0wss@P' |sha1sum
16e89a9108ea2b5cdadf371f8e00868ae67f6074 -
root@JackLinux:~# echo -n '321dr0wss@P' |sha256sum
6671bd249e3b5b4f26c558864c334e5b42b14d80660b8b29a3d707ee0c731031 -
root@JackLinux:~# █

```

```

root@JackLinux:~# echo -n '321dr0wss@PP@ssw0rd123' |md5sum
6045d83ebd7be9f9f7e089a79952c411 -
root@JackLinux:~# echo -n '321dr0wss@PP@ssw0rd123' |sha1sum
0c681e374b06c57df808cc5c2fa6563fec6f2872 -
root@JackLinux:~# echo -n '321dr0wss@PP@ssw0rd123' |sha256sum
0c517f8eff8dd9ba77c29bdcd158c04ca62f703ad849237b74243d1df6dff30e -
root@JackLinux:~# █

```

4b. Linux Summer2024! Output:

```

root@JackLinux:~# echo -n 'Summer2024!' |md5sum
f065d609e55983bc6087c073c91c9bc7 -
root@JackLinux:~# echo -n 'Summer2024!' |sha1sum
7e8b0a3433f1210a9699d85420e363a1b162ecac -
root@JackLinux:~# echo -n 'Summer2024!' |sha256sum
323725e8eff4df0a4974d6ea8c73017aa6467d94e09382745b7a988cec0fba0a -
root@JackLinux:~# echo -n 'Summer2024!' |rev
!4202remmuSroot@JackLinux:~#
root@JackLinux:~# echo -n '!400202remmuS' |md5sum
7c3f3eeda184f5080b6e1bf03914abe7 -
root@JackLinux:~# echo -n '!400202remmuS' |sha1sum
b89697d1fd271374493123c474d1266f45b3a2e9 -
root@JackLinux:~# echo -n '!400202remmuS' |sha256sum
d279c9e6c2004bfbc5f33f32f5f04c42b8a3f2b2b8b3a0b13fec990843f60f1b -
root@JackLinux:~# █

```

```

root@JackLinux:~# echo -n '!4202remmuSSummer2024!' |md5sum
df03ba45bfae63350beda7e8b6db0211 -
root@JackLinux:~# echo -n '!4202remmuSSummer2024!' |sha1sum
93455115d1b62c828c35c2f63afe6f2dfd0ff838 -
root@JackLinux:~# echo -n '!4202remmuSSummer2024!' |sha256sum
54c9aff5ce51068356142e709c098f8ae3b364e7e9b3a3a6701aec6f78acf513 -
root@JackLinux:~# █

```

4c. Linux letmein! Output:

```
root@JackLinux:~# echo -n 'letmein!' |md5sum
2a5de0f53b1317f7e36afcdb6b5202a4 -
root@JackLinux:~# echo -n 'letmein!' |sha1sum
403e35a2b0243d40400af6bb358b5c546cddd981 -
root@JackLinux:~# echo -n 'letmein!' |sha256sum
6a07a5c18e0f6c695c2445e84751f8ac9406da91fae839b5c3de36b9c712269e -
root@JackLinux:~# echo -n 'letmein!' |rev
!niemtelroot@JackLinux:~# echo -n '!niemtel' |md5
Command 'md5' not found, did you mean:
  command 'mdi' from deb linuxcnc-ospace (2.9.1-2ubuntu1)
  command 'cd5' from deb cd5 (0.1-4)
  command 'mdu' from deb mtools (4.0.43-1)
  command 'mdp' from deb mdp (1.0.15-1)
  command 'mda' from deb mailutils-mda (1:3.16-1build1)
  command 'mdl' from deb markdownlint (0.13.0-4)
Try: apt install <deb name>
root@JackLinux:~# echo -n '!niemtel' |md5sum
ec3221c63514fa708d64429ead533444 -
root@JackLinux:~# echo -n '!niemtel' |sha1sum
aed1f2c8014d7008e2d1da6eeba69f22986aa2fc -
root@JackLinux:~# echo -n '!niemtel' |sha256sum
4125aab18ea86020a509c462e422e98eb4dcd4755cfd615eb0930d906dec244a -
root@JackLinux:~#
```

```
root@JackLinux:~# echo -n '!!niemtelletmein!' |md5sum
bacc299caf4c546c6b3f4f5e2b5dc2e8 -
root@JackLinux:~# echo -n '!!niemtelletmein!' |sha1sum
958e918186b48eccc9a91603adf12bb52243a27 -
root@JackLinux:~# echo -n '!!niemtelletmein!' |sha256sum
b54e90d8da0f88cce3f327b0c545797aa78e13248de43be43ae225b62b7298ef -
root@JackLinux:~#
```

Part B — Crack unsalted hashes (CrackStation or john)

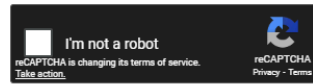
1) P@ssw0rd123 Results:

1a. MD5 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

ccf5538dc31d435d6bab145c924041d8



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
ccf5538dc31d435d6bab145c924041d8	md5	P@ssw0rd123

Color Codes: **Green** Exact match, **Yellow** Partial match, **Red** Not found.

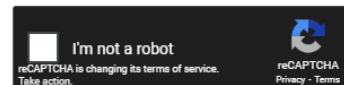
[Download CrackStation's Wordlist](#)

1b. Sha1 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

0f0d959bca569bf2b0a8bff3e2f1e88920ee7c5f



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
0f0d959bca569bf2b0a8bff3e2f1e88920ee7c5f	sha1	P@ssw0rd123

Color Codes: **Green** Exact match, **Yellow** Partial match, **Red** Not found.

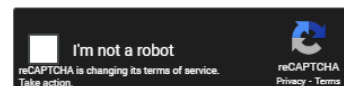
[Download CrackStation's Wordlist](#)

1c. Sha256 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

231ecc7d178da5f22983bc579599396d6c139a457987ae1ee0026d88432d6a72



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1(sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
231ecc7d178da5f22983bc579599396d6c139a457987ae1ee0026d88432d6a72	sha256	P@ssw0rd123

Color Codes: **Green** Exact match, **Yellow** Partial match, **Red** Not found.

[Download CrackStation's Wordlist](#)

1d. Salted Hash MD5, Sha1, Sha256

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

0c517f8eff8dd9ba77c29bdc158c04ca62f703ad849237b74243d1df6dff30e
0c681e374b06c57df808cc5c2fa6563fec6f2872
6045d83ebd7be9f9f7e089a79952c411

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Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
0c517f8eff8dd9ba77c29bdc158c04ca62f703ad849237b74243d1df6dff30e	Unknown	Not found.
0c681e374b06c57df808cc5c2fa6563fec6f2872	Unknown	Not found.
6045d83ebd7be9f9f7e089a79952c411	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

2) Summer2024! Results:

2a. MD5 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

f065d609e55983bc6087c073c91c9bc7

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Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
f065d609e55983bc6087c073c91c9bc7	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

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2b. Sha1 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

7e8b0a3433f1210a9699d85420e363a1b162ecac



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
7e8b0a3433f1210a9699d85420e363a1b162ecac	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

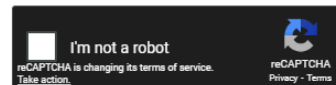
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2c. Sha256 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

323725e8eff4df0a4974d6ea8c73017aa6467d94e09382745b7a988cec0fba0a



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
323725e8eff4df0a4974d6ea8c73017aa6467d94e09382745b7a988cec0fba0a	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

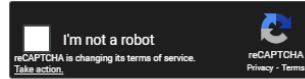
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2d. Salted Hash MD5, Sha1, Sha256

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

```
ee8e9908af71fec9fad6a6f188eba1981dd5209d5f32ab676f16153dafa70338
4efcd01dca161d302a934158c4183725
51fc4501f74a09bfff58f2f4c6d697d5df514068d
```



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, rpeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
ee8e9908af71fec9fad6a6f188eba1981dd5209d5f32ab676f16153dafa70338	Unknown	Not found.
4efcd01dca161d302a934158c4183725	Unknown	Not found.
51fc4501f74a09bfff58f2f4c6d697d5df514068d	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

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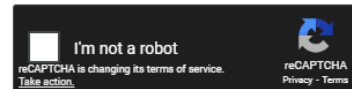
3) letmein! Results:

3a. MD5 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

```
2a5de0f53b1317f7e36afcdb6b5202a4
```



Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, rpeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin), QubesV3.1BackupDefaults

Hash	Type	Result
2a5de0f53b1317f7e36afcdb6b5202a4	md5	letmein!

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

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3b. Sha1 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

403e35a2b0243d40400af6bb358b5c546cddd981



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
403e35a2b0243d40400af6bb358b5c546cddd981	sha1	letmein!

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

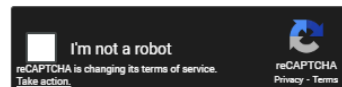
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3c. Sha256 Unsalted Hash

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

6a07a5c18e0f6c695c2445e84751f8ac9406da91fae839b5c3de36b9c712269e



Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
6a07a5c18e0f6c695c2445e84751f8ac9406da91fae839b5c3de36b9c712269e	sha256	letmein!

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

[Download CrackStation's Wordlist](#)

3d. Salted Hash MD5, Sha1, Sha256

Free Password Hash Cracker

Enter up to 20 non-salted hashes, one per line:

ae2698eef616d5de57c77f933d296a9f54373702054704ae343e815416957aa5
a896024e0fe39b965eaa4c02df4e1e55
b47b4745d3b3363efa211fc05d687ea260c5e66a



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Crack Hashes

Supports: LM, NTLM, md2, md4, md5, md5(md5_hex), md5-half, sha1, sha224, sha256, sha384, sha512, ripeMD160, whirlpool, MySQL 4.1+ (sha1 sha1_bin)), QubesV3.1BackupDefaults

Hash	Type	Result
ae2698eef616d5de57c77f933d296a9f54373702054704ae343e815416957aa5	Unknown	Not found.
a896024e0fe39b965eaa4c02df4e1e55	Unknown	Not found.
b47b4745d3b3363efa211fc05d687ea260c5e66a	Unknown	Not found.

Color Codes: Green Exact match, Yellow Partial match, Red Not found.

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